

WTW 158 2018 Worksheet 4 on Inverse Trig functions and Limits.

A. TEXTBOOK PROBLEMS

p66 no 63, 64, 68, 70, 71, 72, 75 p 82 no 1(a), 6 (a) - (i)

p 92 no 7, 9, 11, 17, 31, 34, 38

B. MORE PROBLEMS

1.1 Complete the definition:

$y = \cos x \Leftrightarrow \dots$ where $\dots$

1.2 Sketch $y = \cos x, x \in [0, \pi]$ and $y = \arccos x$ on the same set of axes

2. Find the domain of the function $f(x) = \frac{1}{\arcsin(x)}$

3. Find the average rate of change of the function $f(x) = \ln x$ over the interval $[1, 3]$

4. Does the function $f(x) = \frac{x^2+2x-8}{x^2-5x+6}$ have a vertical asymptote(s)?

5. Determine $\lim_{x \rightarrow 0} (\frac{1}{x} - \ln x)$

6. (i) $\lim_{x \rightarrow 2^+} \frac{x}{2-x}$ (ii) $\lim_{x \rightarrow 2^-} \frac{x+1}{x-2}$

C. TEST QUESTIONS

Question 1

For which value(s) of x is $\tan(\tan^{-1}x) = x$?

(Notation $\tan^{-1} = \arctan = \text{bg tan}$)

(a)	$0 \leq x \leq \pi$	(b)	$-1 \leq x \leq 1$	(c)	$-\pi \leq x \leq \pi$
(d)	$-\frac{\pi}{2} < x < \frac{\pi}{2}$	(e)	$x \in \mathbb{R}$	(f)	None of these

Question 2

$\lim_{x \rightarrow a} \frac{x^2 - 2ax + a^2}{x^2 - a^2} =$

(a)	$\frac{1}{2a}$	(b)	$-\frac{1}{2a}$	(c)	$-\frac{1}{2a^2}$
(d)	$\frac{1}{2a^2}$	(e)	0	(f)	None of these

Question 3

Does the function $f(x) = \frac{|x+5|}{x+5}$ have a vertical asymptote at $x = -5$? Explain.

D YOUTUBE

Kahn Academy - Introduction to limits

SELECTED ANSWERS

A. p66 63. π 64. $\frac{\pi}{3}$ 68. $-\frac{\pi}{4}$ 70. $\frac{x}{\sqrt{1-x^2}}$ 71. $\frac{x}{\sqrt{1+x^2}}$ 72. $2x\sqrt{1-x^2}$ 75. Domain: $[-\frac{2}{3}, 0]$

Range: $[-\frac{\pi}{2}, \frac{\pi}{2}]$ p82 6(a) (i) 4.42 m/s p92 7(a) -1 (b) -2 (c) DNE (d) 2 (e) 0 (f) DNE

(g) 1 (h) 3 9. (a) $-\infty$ (b) ∞ (c) ∞ (d) $-\infty$ (e) ∞ 11. $\lim_{x \rightarrow a} f(x)$ exists for $a \in \mathbb{R} \setminus \{-1\}$ 31. ∞

34. $-\infty$ 38. $-\infty$

B. 1.1 See textbook 2. $x \in [-1, 1] \setminus \{0\}$ 3. $\frac{\ln 3}{2}$ 4. VA $x = 3$ 5. ∞ 6. (i) $-\infty$ (ii) $-\infty$

C. 1 (e) 2(e) 3. No, $\lim_{x \rightarrow 5^+} f(x) = 1$, $\lim_{x \rightarrow 5^-} f(x) = -1$